Cheeseproducers

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R Markdown

TWO WAY ANOVA: with/without interaction

We want to study the rendibility of cheese producers.

Three types of milk are considered.

```
library(car)
```

```
## Loading required package: carData
```

```
library(HH)
```

```
## Loading required package: lattice
## Loading required package: grid
## Loading required package: latticeExtra
## Loading required package: RColorBrewer
## Loading required package: multcomp
## Loading required package: mvtnorm
## Loading required package: survival
## Loading required package: TH.data
## Loading required package: MASS
##
## Attaching package: 'TH.data'
## The following object is masked from 'package:MASS':
##
##     geyser
## Loading required package: gridExtra
##
## Attaching package: 'HH'
## The following objects are masked from 'package:car':
##
##     logit, vif
```

```
library(car)
library(emmeans)
```

```
##
## Attaching package: 'emmeans'
```

```
## The following object is masked from 'package:HH':
##
##     as.glht
## The following object is masked from 'package:multcomp':
##
##     cld
```

```
library(tables)
```

```
## Loading required package: Hmisc
## Loading required package: Formula
## Loading required package: ggplot2
##
## Attaching package: 'ggplot2'
## The following object is masked from 'package:latticeExtra':
##
##     layer
##
## Attaching package: 'Hmisc'
## The following objects are masked from 'package:base':
##
##     format.pval, units
```

```
library(RcmdrMisc)
```

```
## Loading required package: sandwich
##
## Attaching package: 'RcmdrMisc'
## The following object is masked from 'package:Hmisc':
##
##     Dotplot
```

Loading the data and printing the top part

```
setwd("G:/PiE2/2018")
cheesedata<-read.csv2("./Dades/cheese.csv")
head(cheesedata)
```

```
##   THERMIC CaCl2    COW   GOAT  SHEEP
## 1   plane   yes 46.960 50.565 53.345
## 2   plane   yes 37.235 37.685 53.790
## 3   plane   yes 36.205 34.620 52.635
## 4   plane   yes 38.835 40.920 50.330
## 5   plane    no 61.120 44.790 73.530
## 6   plane    no 53.570 37.015 57.770
```

```
dim(cheesedata)
```

```
## [1] 16  5
```

As it can be seen, there are a total of 16 observations for each one of the milk types. For a given type of milk, one has four observations for each combination of thermic treatment (2 levels) and CaC12 additive (2 levels). To confirm that we compute the following table:

```
tabular((COW+GOAT+SHEEP)*(THERMIC+1)~(CaCl2+1)*(n=1),cheesedata)
```

		CaCl2		
		no	yes	All
	THERMIC	n	n	n
COW	pasteurized	4	4	8
	plane	4	4	8
	All	8	8	16
GOAT	pasteurized	4	4	8
	plane	4	4	8
	All	8	8	16
SHEEP	pasteurized	4	4	8
	plane	4	4	8
	All	8	8	16

Let us start by doing the descriptive statistics

```
tabular((COW+GOAT+SHEEP)*(THERMIC+1)~(CaCl2+1)*mean,cheesedata)
```

		CaCl2		
		no	yes	All
	THERMIC	mean	mean	mean
COW	pasteurized	50.54	46.00	48.27
	plane	58.56	39.81	49.19
	All	54.55	42.90	48.73
GOAT	pasteurized	49.57	43.23	46.40
	plane	42.35	40.95	41.65
	All	45.96	42.09	44.02
SHEEP	pasteurized	72.30	65.48	68.89
	plane	66.64	52.52	59.58
	All	69.47	59.00	64.24

Looking at the means table we see that:

1) For each one of the milks, if the additive is present, the rendibility is smaller, independent of if the milk is pasteurized or plane.

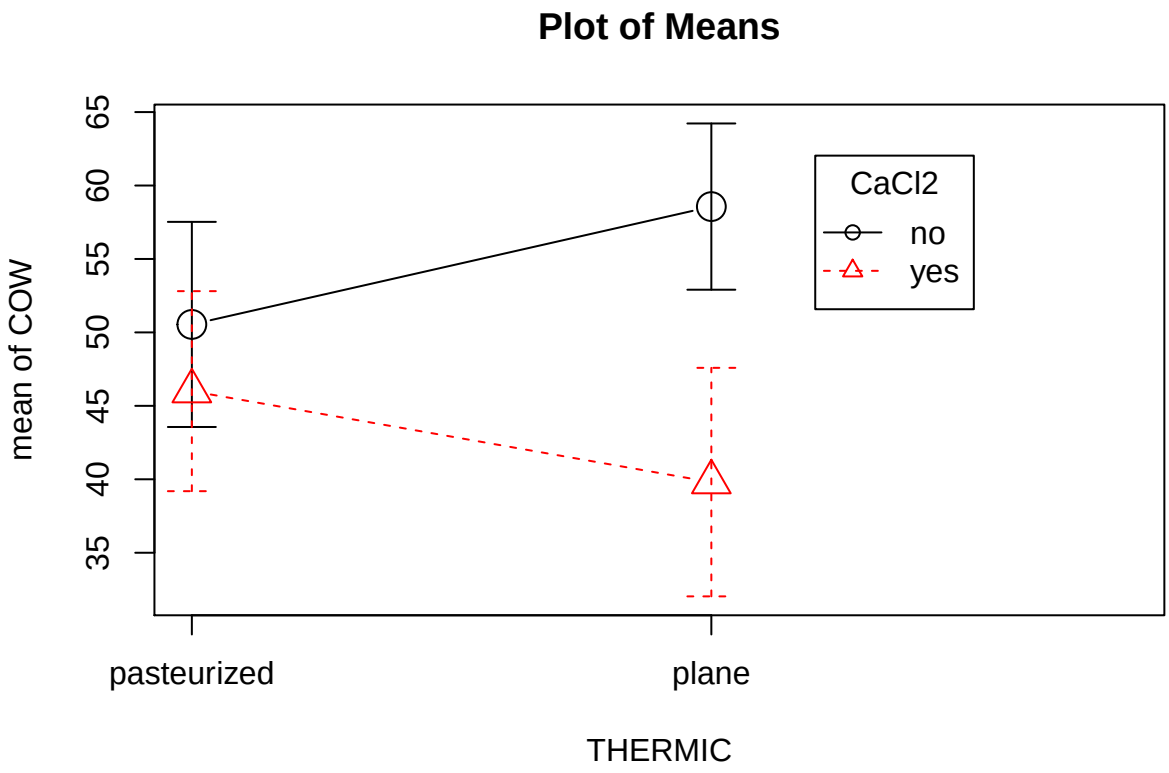
2) For COW milk, to use pasteurized milk reduces the rendibility, for the other milk types it is the other way around.

```
tabular((COW+GOAT+SHEEP)*(THERMIC)~(CaCl2)*sd,cheesedata)
```

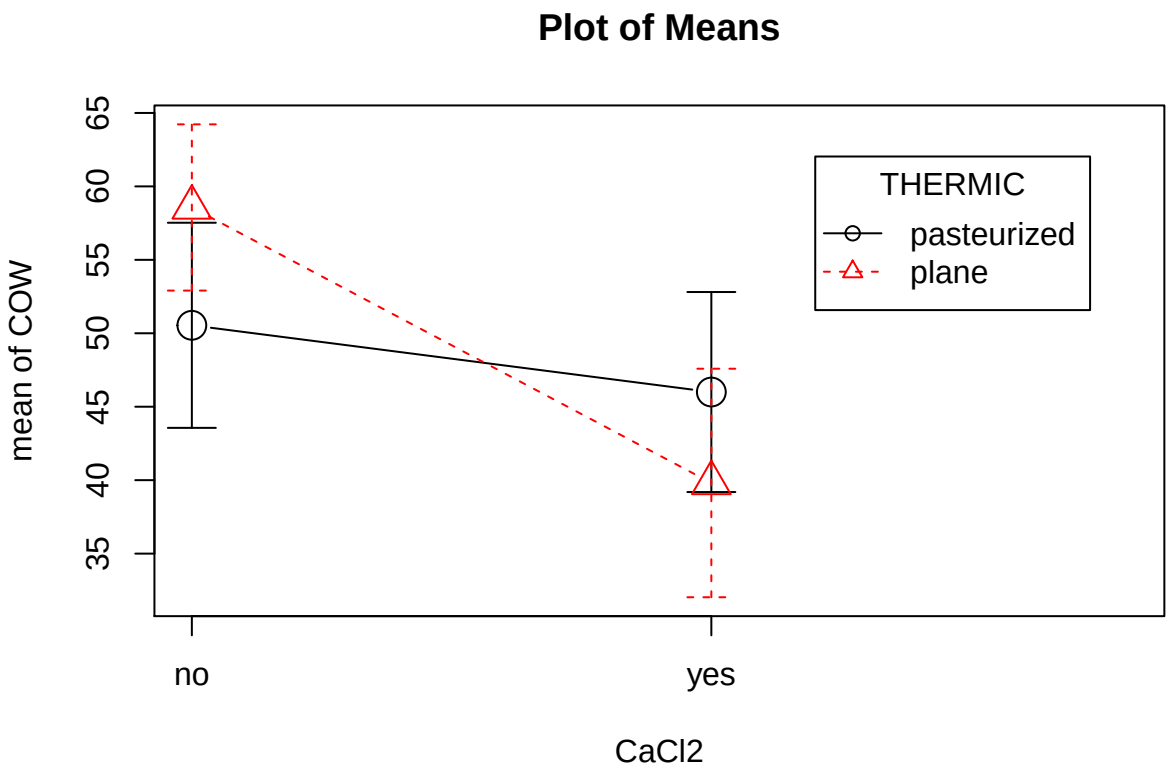
		CaCl2	
		no	yes
	THERMIC	sd	sd
COW	pasteurized	4.388	4.279
	plane	3.556	4.889
GOAT	pasteurized	2.934	2.760
	plane	3.621	6.908
SHEEP	pasteurized	3.396	4.742
	plane	9.767	1.539

Looking at the standard deviation table we see that for the SHEEP and plane milk, there is a large difference between the sds of the additive/not additive groups.

```
with(cheesedata, plotMeans(COW, THERMIC, CaCl2, error.bars="conf.int", level=0.95, connect=TRUE))
```



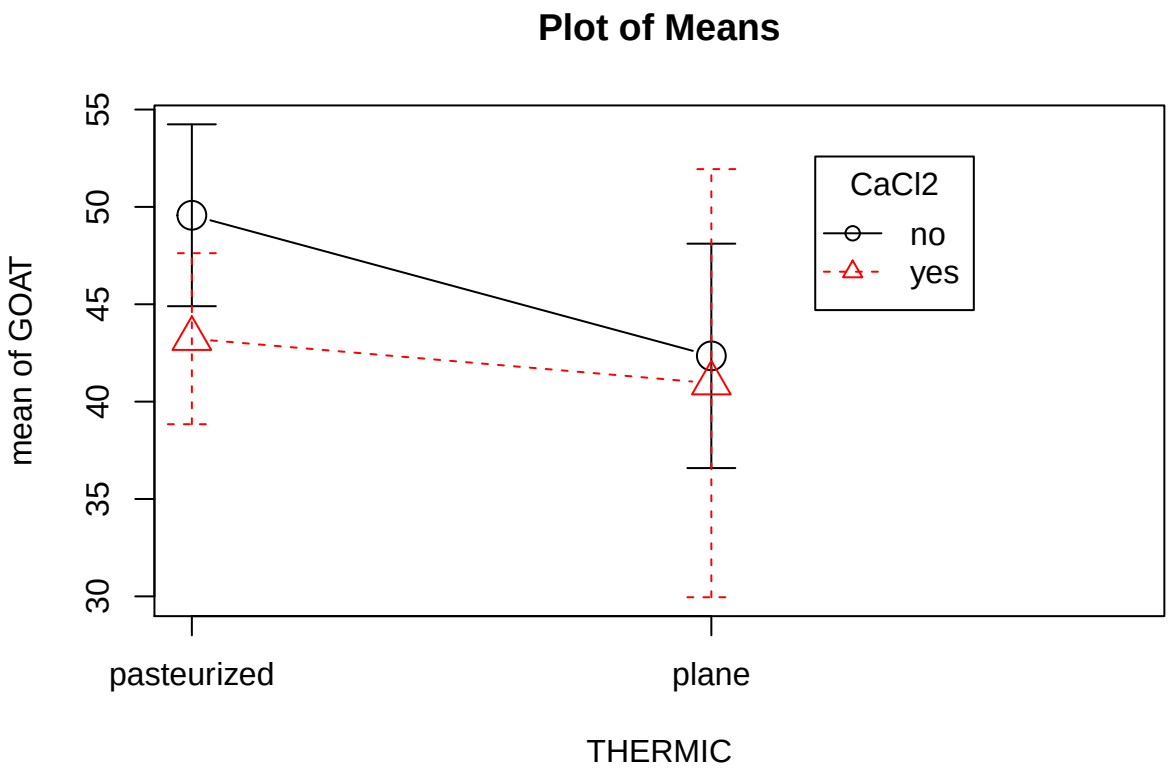

```
with(cheesedata, plotMeans(COW, CaCl2, THERMIC, error.bars="conf.int", level=0.95, connect=TRUE))
```



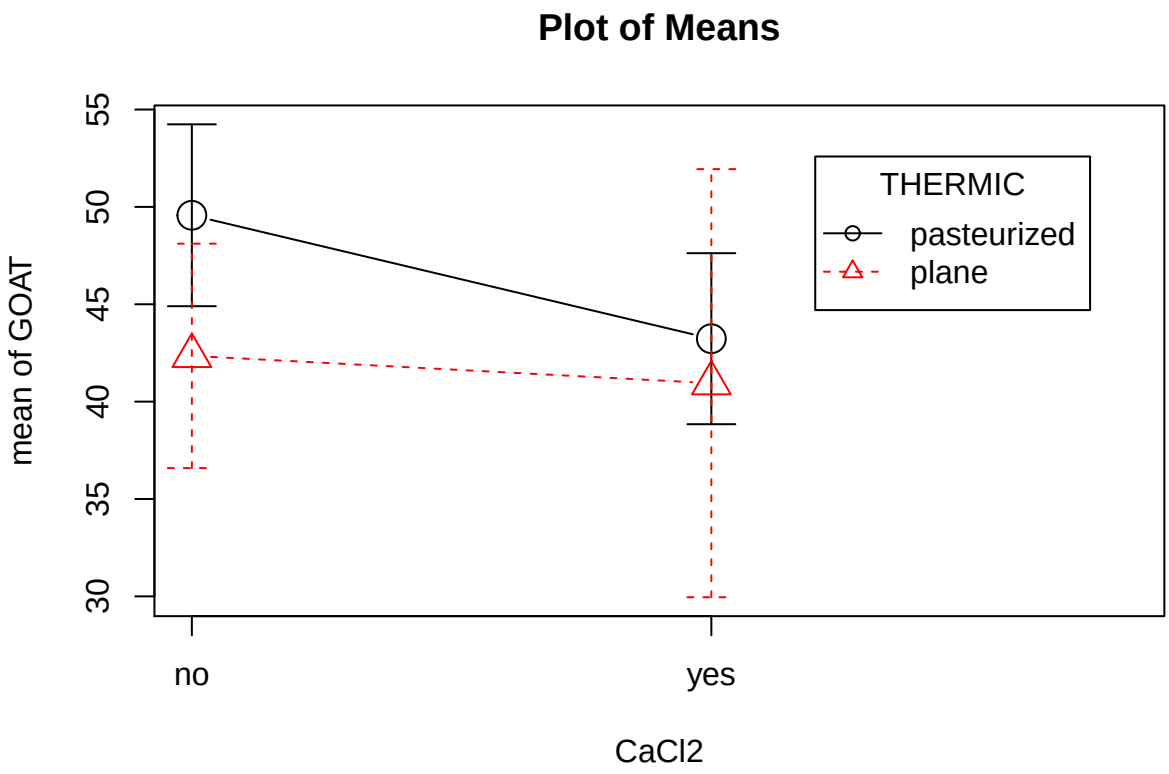
With respect to the plot of the means of the Cow milk, we can say that.

The rendibility obtained with/without the CaC12 seems not to be different if the cheese is made with plane milk, but it seems to be different if the milk is elaborated with pasteurized milk. This tells us that the use of the additive will be more or less convinient depending on the type of milk we are using to produce the cheese. We also see that the rendibility decreases considerably if we add the additive CaC12 if we are working with plane milk. The additive turn out to be less importnat if we work with pasteurized milk. The fact that the polygonals cross each other clearly shows that the factors thermic treatment and CaC12 interact.

```
with(cheesedata, plotMeans(GOAT, THERMIC, CaCl2, error.bars="conf.int", level=0.95, connect=TRUE))
```

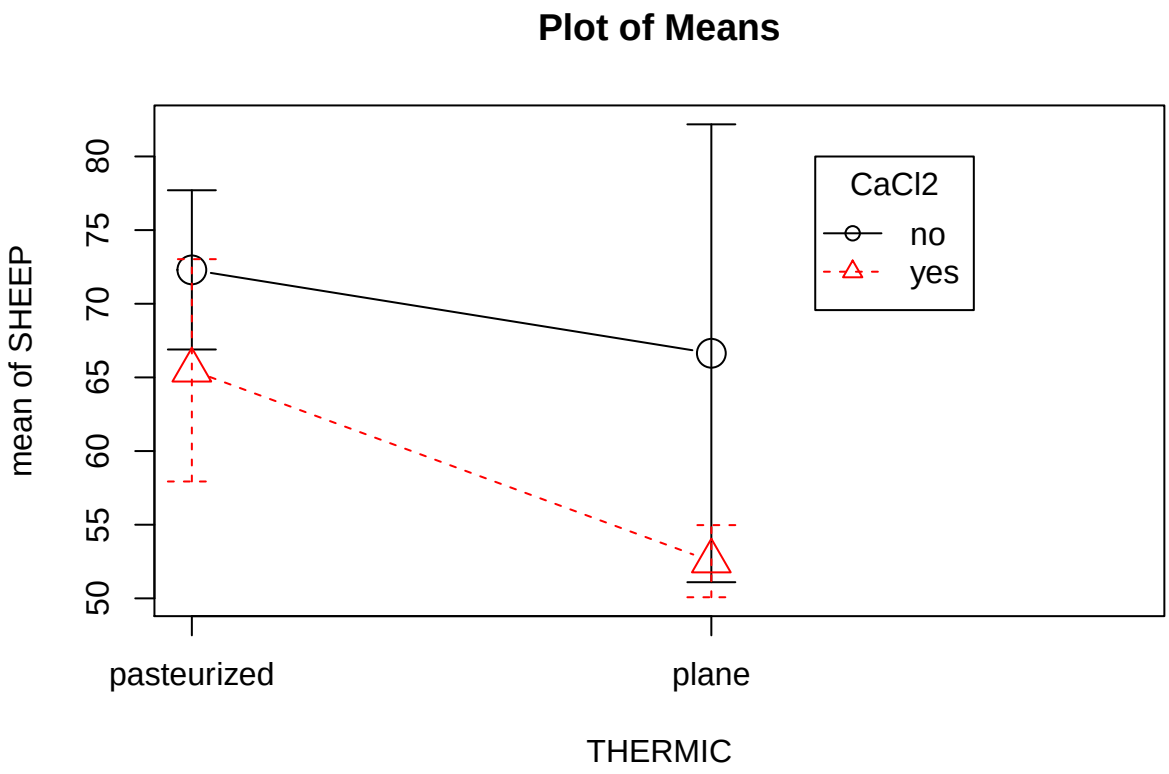



```
with(cheesedata, plotMeans(GOAT, CaCl2, THERMIC, error.bars="conf.int", level=0.95, connect=TRUE))
```

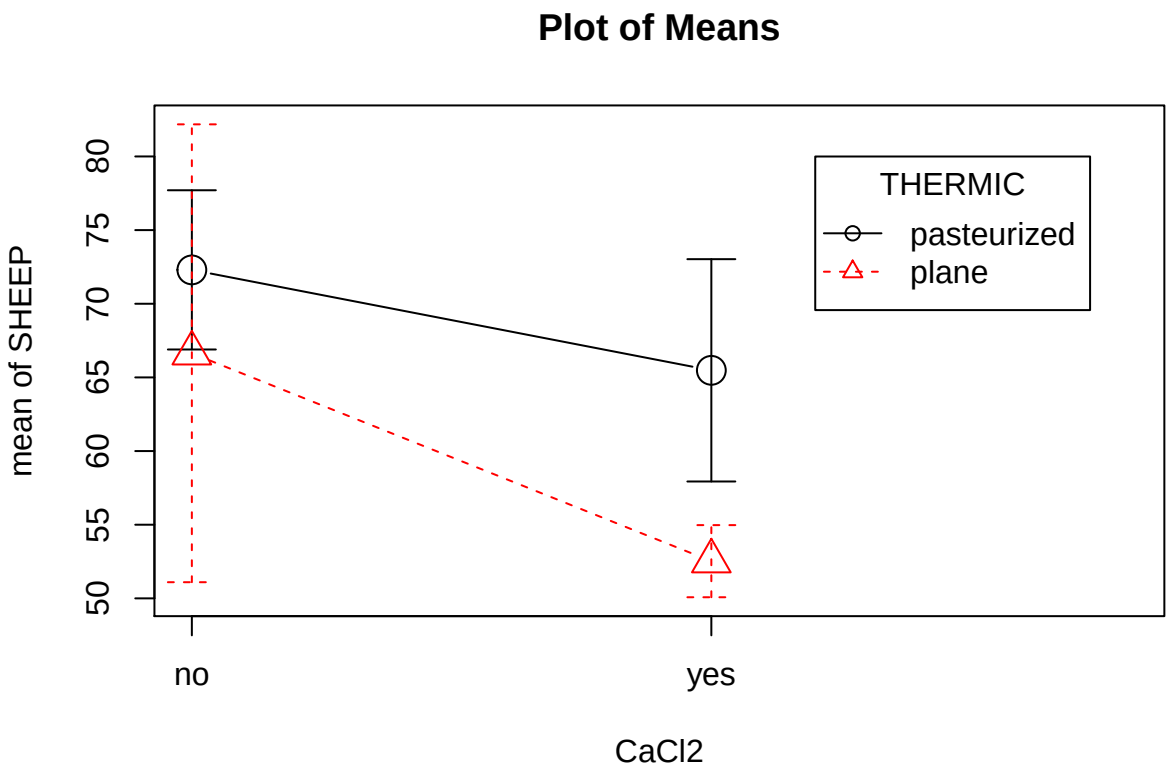


Performing the same plots with the GOAT milk, we see that the rendibility decreases if we use plane milk instead of pasteurized milk, but this decrease dosen't differ much in the group of milk with CaC12 and the group without CaC12. A simlilar behaviour is observed if we interchange the milk with the additive factors.

```
with(cheesedata, plotMeans(SHEEP, THERMIC, CaCl2, error.bars="conf.int", level=0.95, connect=TRUE))
```




```
with(cheesedata, plotMeans(SHEEP, CaCl2, THERMIC, error.bars="conf.int", level=0.95, connect=TRUE))
```



Similarly, the interaction between the two factors is not so clear in the SHEEP milk rendibility. Less rendibility is observed with the milks that contain the additive and the decrease seems to be a little bit large in the plane milk.

COW milk analysis

We first compute the means table

```
tabular(THERMIC~COW*CaCl2*mean,cheesedata)
```

	COW	
	CaCl2	
	no	yes
THERMIC	mean	mean
pasteurized	50.54	46.00
plane	58.56	39.81

We now fit a linear model: TWO-WAR ANOVA with interaction

```
mcow<-lm(COW~THERMIC*CaCl2,cheesedata)
summary(mcow)
```

```
##
## Call:
```

```
## lm(formula = COW ~ THERMIC * CaCl2, data = cheesedata)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.9850 -3.2138  0.1913  2.5437  7.1513
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)             50.543      2.152  23.484 2.12e-11 ***
## THERMICplane             8.023      3.044   2.636  0.02174 *
## CaCl2yes                -4.542      3.044  -1.492  0.16141
## THERMICplane:CaCl2yes  -14.214      4.304  -3.302  0.00632 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.304 on 12 degrees of freedom
## Multiple R-squared:  0.7709, Adjusted R-squared:  0.7136
## F-statistic: 13.46 on 3 and 12 DF,  p-value: 0.0003799
```

In this model we see thath the thermic treatment is significative and also it is the interaction within termic treatment and CaC12. Factor CaC12 is not significant. The two-way model with interaction explains 77% of the variability observed in the rendibility. The null hypothesis of the omnibus test is rejected, meaning that the two factors considered explain an important part of the variability explained in the rendibility.

```
anova(mcow)
```

```
## Analysis of Variance Table
##
## Response: COW
##               Df Sum Sq Mean Sq F value    Pr(>F)
## THERMIC        1   3.35    3.35   0.181 0.6780528
## CaCl2          1 542.83  542.83  29.297 0.0001568 ***
## THERMIC:CaCl2  1 202.03  202.03  10.904 0.0063162 **
## Residuals     12 222.34   18.53
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Anova(mcow,t=3)
```

```
## Anova Table (Type III tests)
##
## Response: COW
##                Sum Sq Df  F value    Pr(>F)
## (Intercept)   10218.2  1 551.4871 2.124e-11 ***
## THERMIC         128.7  1   6.9472  0.021743 *
## CaCl2            41.3  1   2.2273  0.161406
## THERMIC:CaCl2   202.0  1  10.9038  0.006316 **
## Residuals       222.3 12
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In the anova table appears the Thermic factor as not significative and the factor CaC12 as significative. The interaction also truns out to be significative. Nevertheless, these conclusions depend on the order in which the factors are included in the model. That's why the type III sums of squares are computed with the Anova sentence, since these are independent of the order. The results of the Anova tells us that the main effect of CaC12 is not significative while the thermic and interaction components are significant. This is coherent with the summary of the model that has been obtained.

In what follows let us compare the means of the rendibility by the Tukey method. Given that there is interaction, the results obtained for a given factor will depend on the level of the other factor.

```
CLD(emmeans(mcow,~THERMIC|CaCl2),Letters=letters,reversed=T)
```

```
## CaCl2 = no:
##  THERMIC       emmean       SE df lower.CL upper.CL .group
##  plane       58.56500 2.152232 12 53.87569 63.25431  a
##  pasteurized 50.54250 2.152232 12 45.85319 55.23181   b
##
## CaCl2 = yes:
##  THERMIC       emmean       SE df lower.CL upper.CL .group
##  pasteurized 46.00000 2.152232 12 41.31069 50.68931  a
##  plane       39.80875 2.152232 12 35.11944 44.49806  a
##
## Confidence level used: 0.95
## significance level used: alpha = 0.05
```

```
CLD(emmeans(mcow,~CaCl2|THERMIC),Letters=letters,reversed=T)
```

```
## THERMIC = pasteurized:
##  CaCl2   emmean       SE df lower.CL upper.CL .group
##  no    50.54250 2.152232 12 45.85319 55.23181  a
##  yes   46.00000 2.152232 12 41.31069 50.68931  a
##
## THERMIC = plane:
##  CaCl2   emmean       SE df lower.CL upper.CL .group
##  no    58.56500 2.152232 12 53.87569 63.25431  a
##  yes   39.80875 2.152232 12 35.11944 44.49806   b
##
## Confidence level used: 0.95
## significance level used: alpha = 0.05
```

Here we see that:

1) If CaC12 is no (no additive in the milk), there are differences in rendibility due to the milk. If CaC12 is yes, then the rendibility of the two milks are not statistically different.

2) If we decide to use pasteurized milk, then we do not have to worry about the additive, because the rendibility is statistically the same.

3) If we decide to use plane milk, then we will obtain a larger rendibility if Cac12 is not present.

Next table summarizes the results obtained previously.

```
CLD(emmeans(mcow,~THERMIC*CaCl2),Letters=letters,reversed=T)
```

```
##  THERMIC     CaCl2   emmean       SE df lower.CL upper.CL .group
##  plane       no    58.56500 2.152232 12 53.87569 63.25431  a
##  pasteurized no    50.54250 2.152232 12 45.85319 55.23181  ab
##  pasteurized yes   46.00000 2.152232 12 41.31069 50.68931   bc
##  plane       yes   39.80875 2.152232 12 35.11944 44.49806    c
##
## Confidence level used: 0.95
## P value adjustment: tukey method for comparing a family of 4 estimates
## significance level used: alpha = 0.05
```

Observe in what follows that if we do not fix the level of one factor previously to perfomr the comparison, R will warning us about the presence of interaction that we are not taking into account.

```
CLD(emmeans(mcow,~CaCl2),Letters=letters,reversed=T)
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
##  CaCl2   emmean       SE df lower.CL upper.CL .group
##  no    54.55375 1.521858 12 51.23791 57.86959  a
##  yes   42.90438 1.521858 12 39.58853 46.22022   b
##
## Results are averaged over the levels of: THERMIC
## Confidence level used: 0.95
## significance level used: alpha = 0.05
```

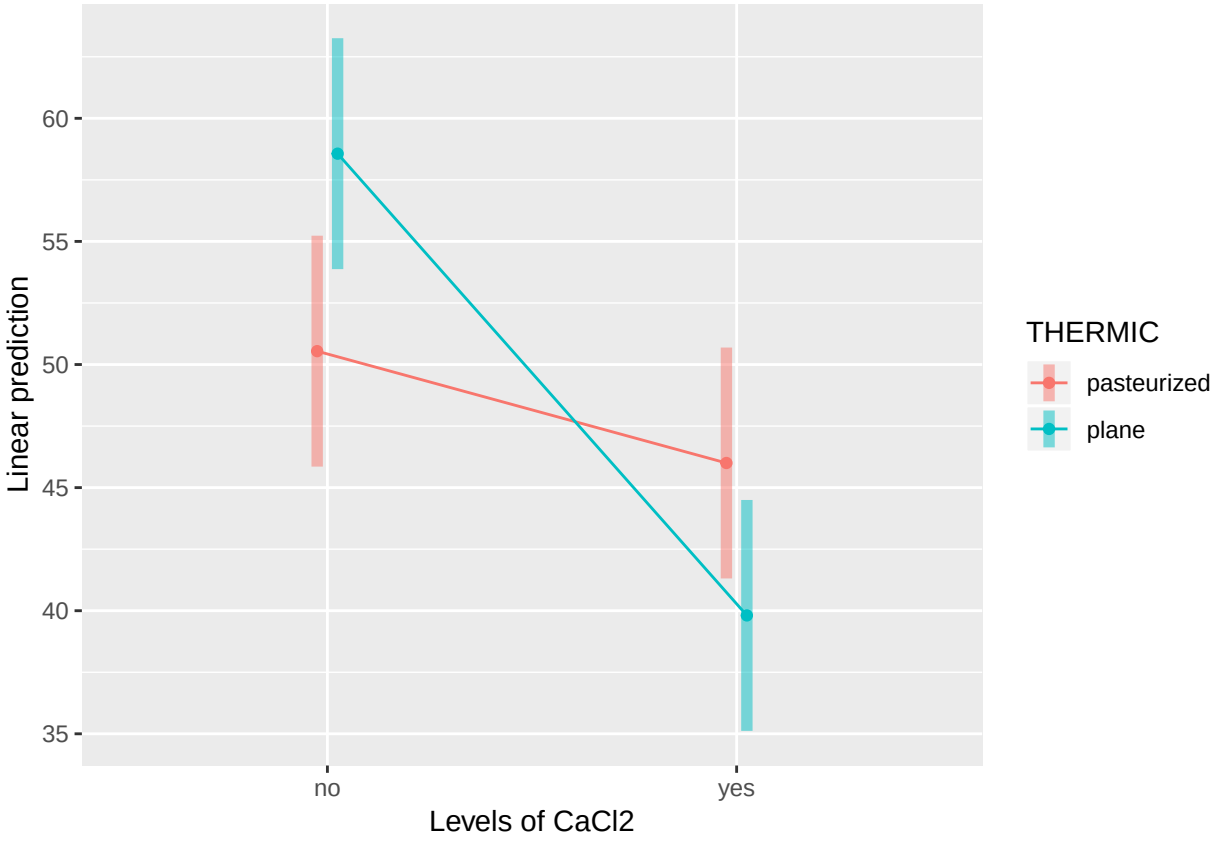
```
CLD(emmeans(mcow,~THERMIC),Letters=letters,reversed=T)
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

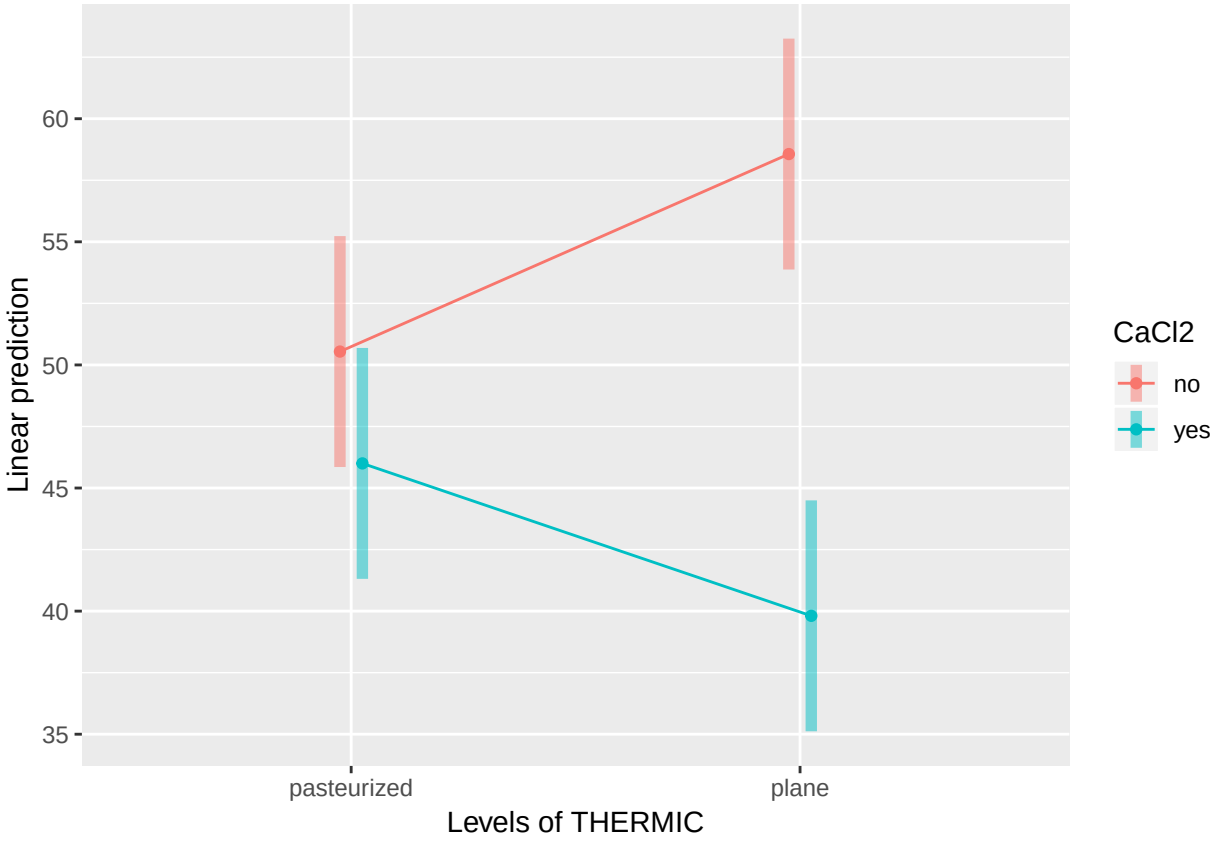
```
##  THERMIC       emmean       SE df lower.CL upper.CL .group
##  plane       49.18688 1.521858 12 45.87103 52.50272  a
##  pasteurized 48.27125 1.521858 12 44.95541 51.58709  a
##
## Results are averaged over the levels of: CaCl2
## Confidence level used: 0.95
## significance level used: alpha = 0.05
```

In what follows it appears the predicted rendibility obtained with our model jointtly with the confidence intervals for the predicted means.

```
(emmip(mcow,THERMIC~CaCl2,CIs=TRUE))
```



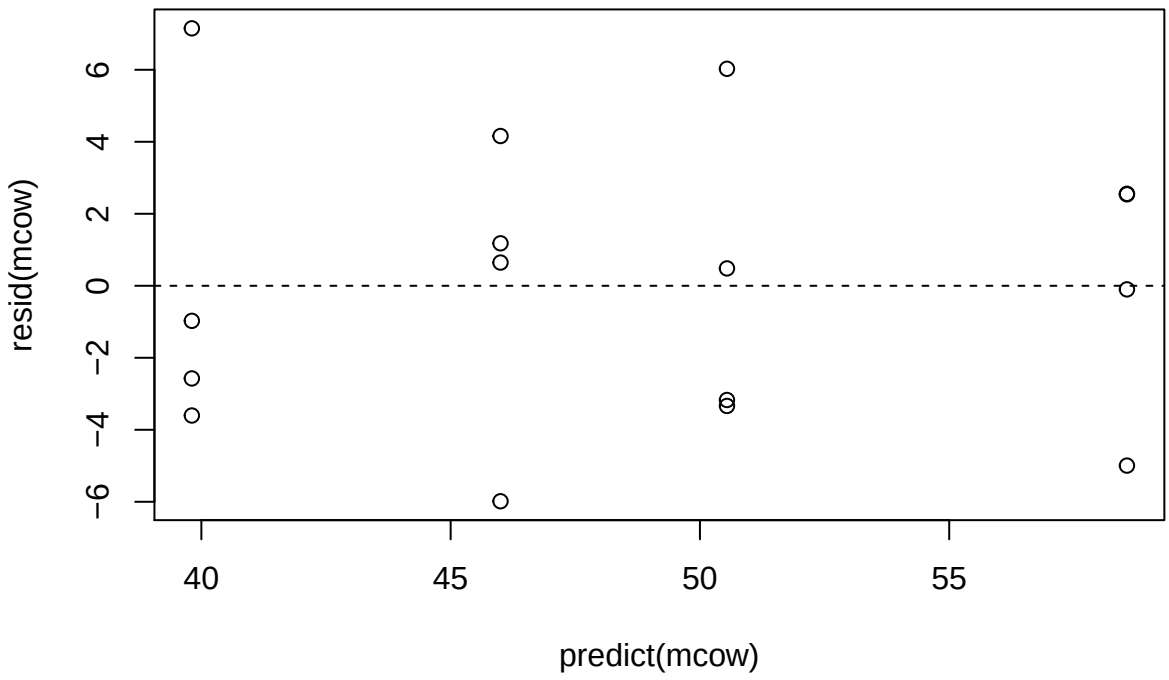
```
(emmip(mcow,CaCl2~THERMIC,CIs=TRUE))
```



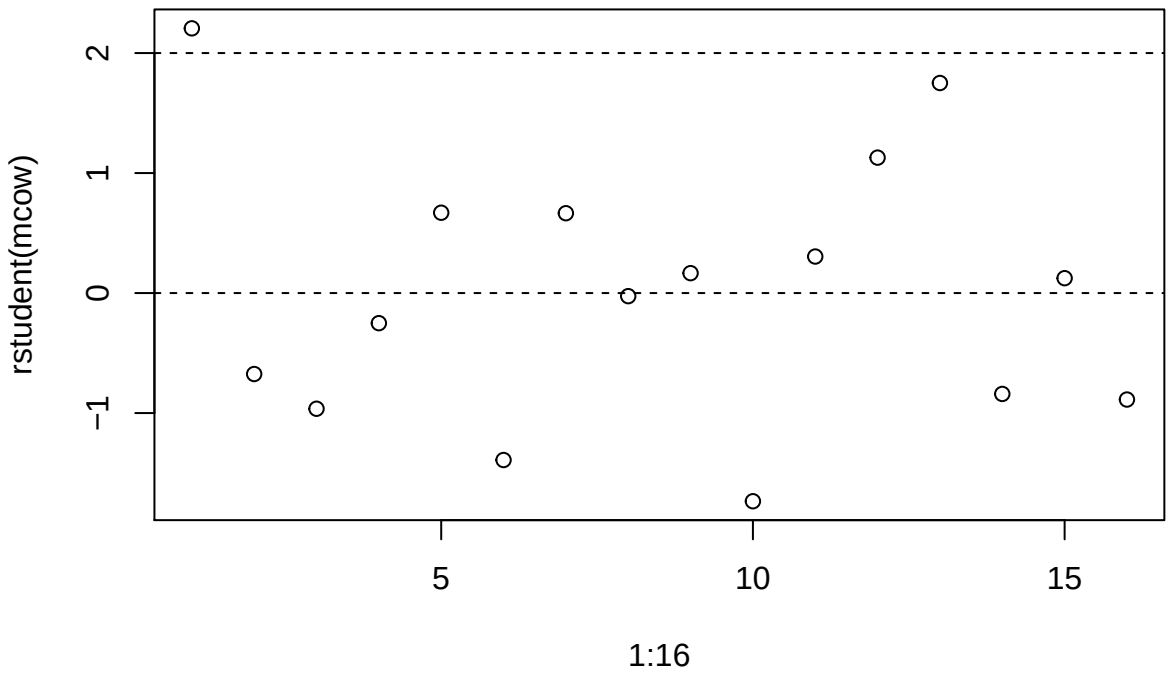
In what follows we perform the residual analysis

In order to see if we can accept the hypothesis of the linear model, we perform the residual analysis.

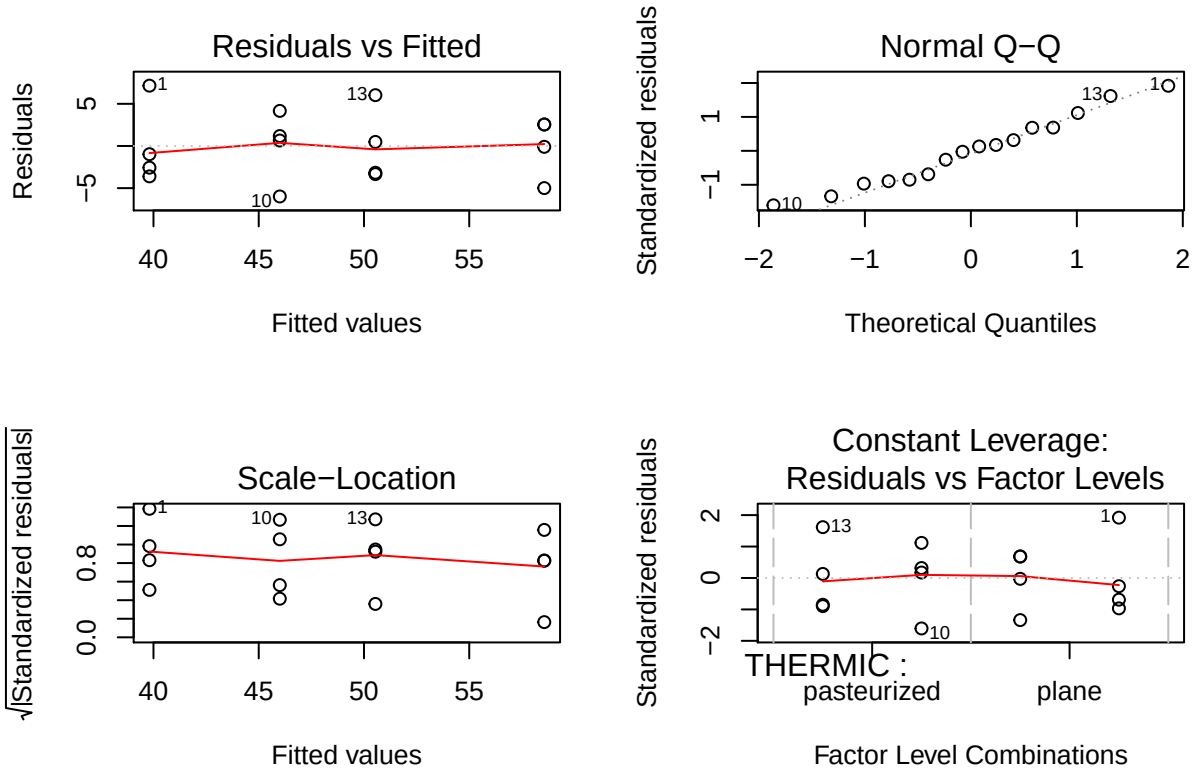
```
plot(predict(mcow),resid(mcow))
abline(h=0,lty=2)
```




```
plot(1:16,rstudent(mcow))
abline(h=c(-2,0,2),lty=2)
```




```
oldpar <- par( mfrow=c(2,2))
plot(mcow,ask=F)
```




```
par(oldpar)
```

We have four predicted values, one for each cell 8combination of thermic and CaC12). In the residual versus predicted plot we do not observe any pattern and we clearly see thath the residual variance is approximatly the same in each cell, so the homocedasticity property seems to be correct.

The studentized residuals also do not show any pattern, and observations with big studentized residuals are not observed.

In the four in one plot we see that the Normality may also be assumed, since the qq-plot is quite linear. also the independence may be assumed since no patterns are observed.

We leave to the students to perform similar analysis with the other two types of milks.